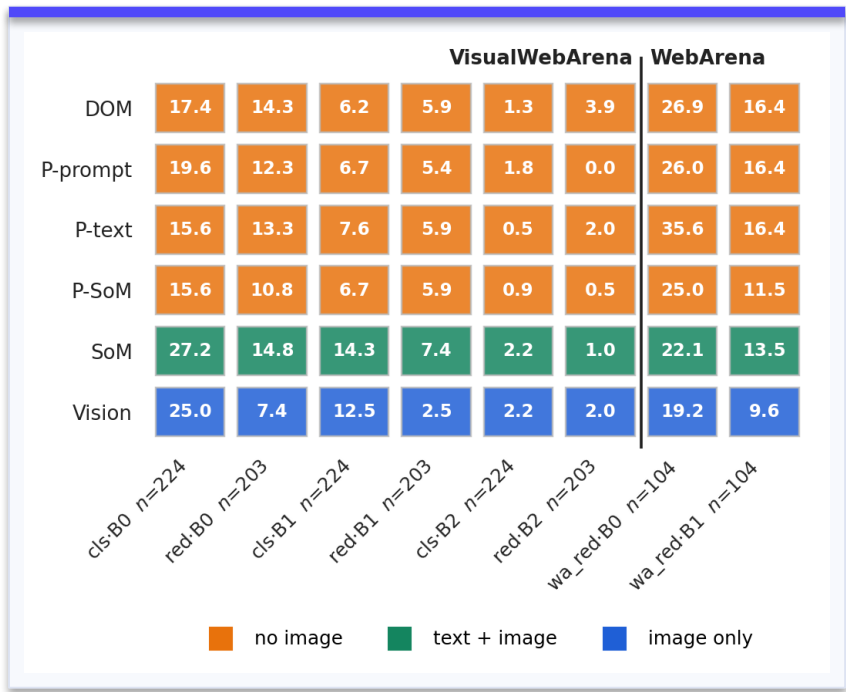
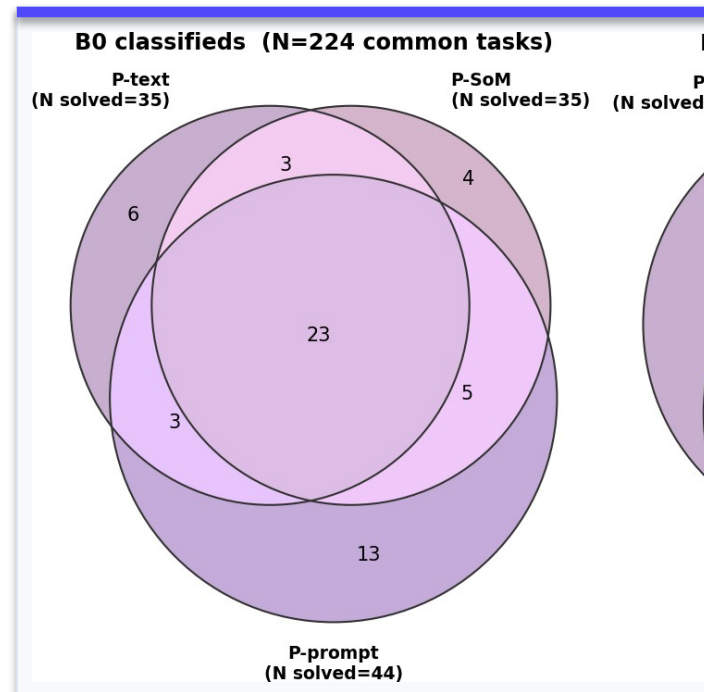


Same start. READ finishes; LOOK loops back. Both outcomes repeat on an independent rerun.

1 · NO VIEW WINS EVERYWHERE

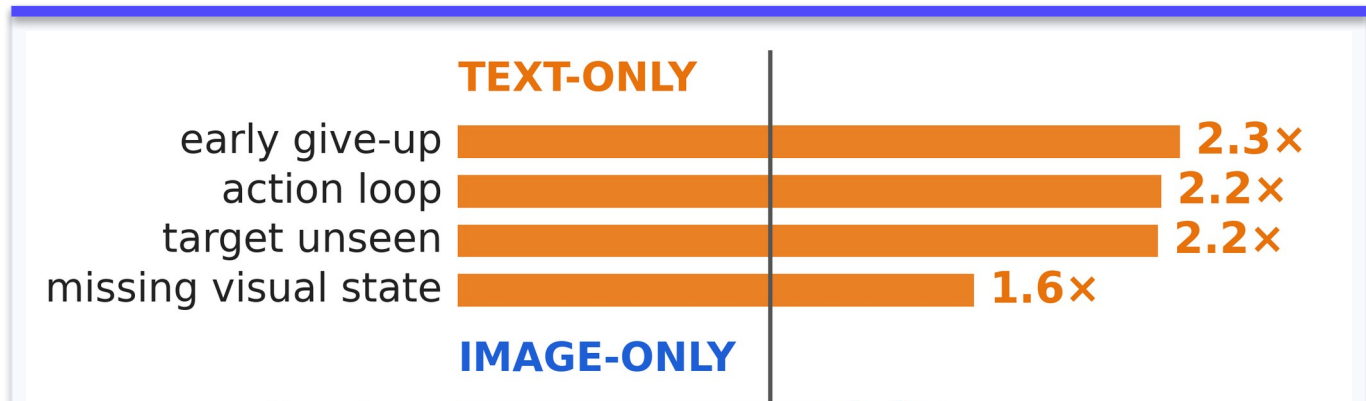


2 · THEY SOLVE DIFFERENT

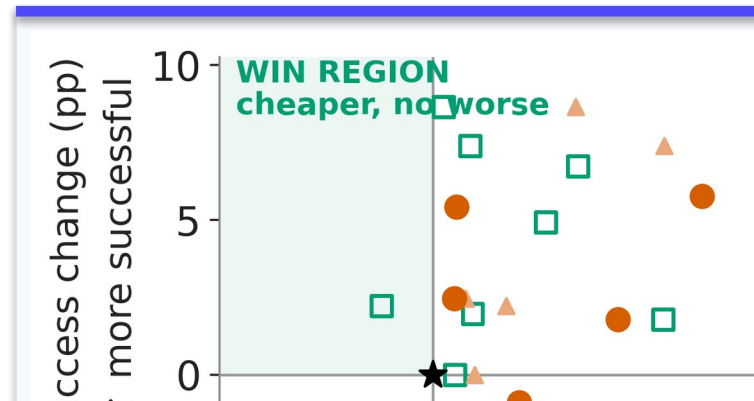


The sets overlap — but do not coin

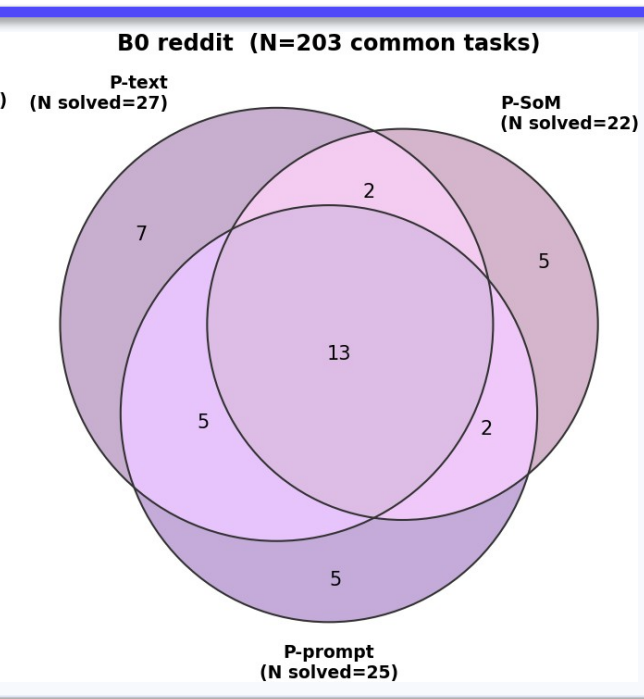
4 · AND THEY FAIL DIFFERENTLY



5 · SO BUILD A ROUTER? NO



DIFFERENT TASKS

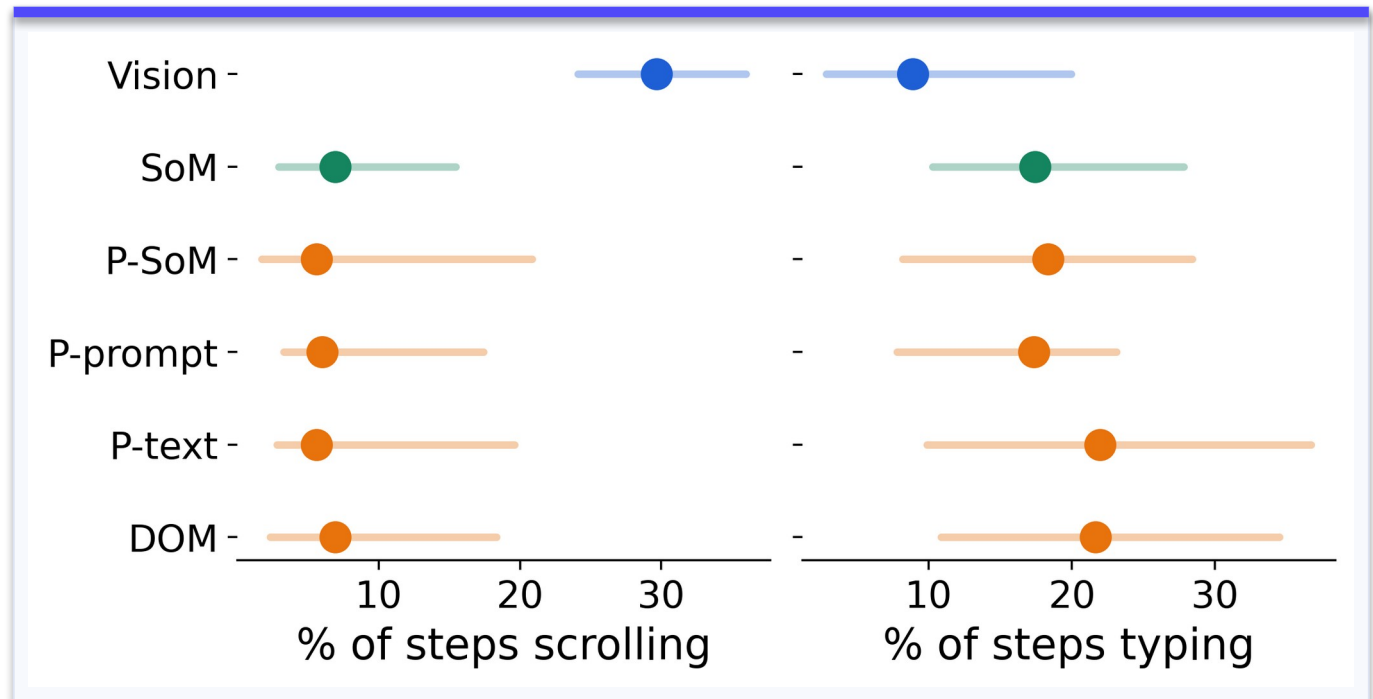


do not coincide.

NOT SO FAST.

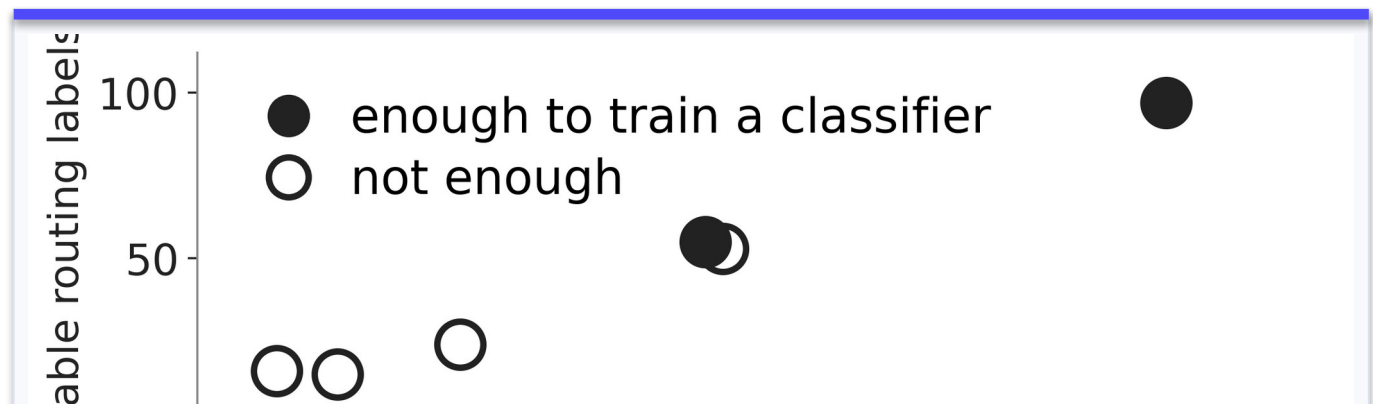


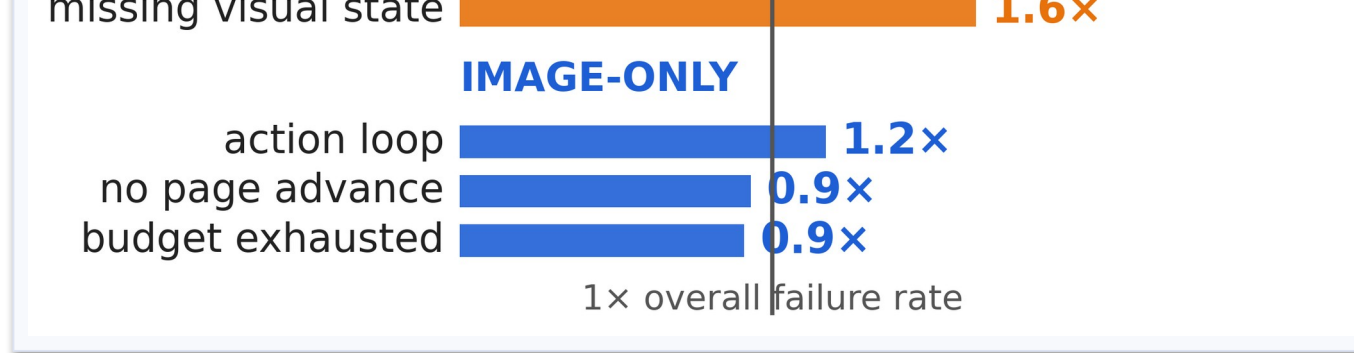
3 · THEY BEHAVE DIFFERENTLY



Vision scrolls ~4× more.

6 · AND THIS IS WHY





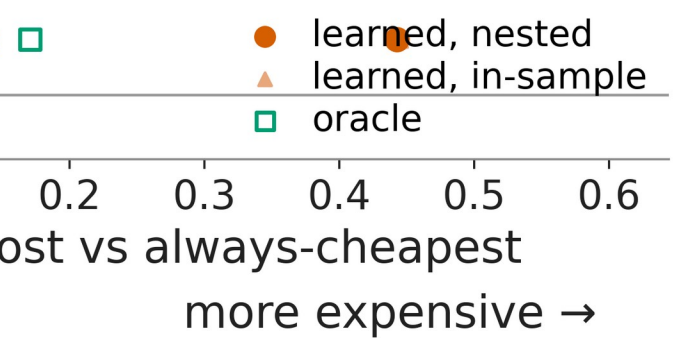
Text-only skews toward early give-up; image-only toward stalled progress.



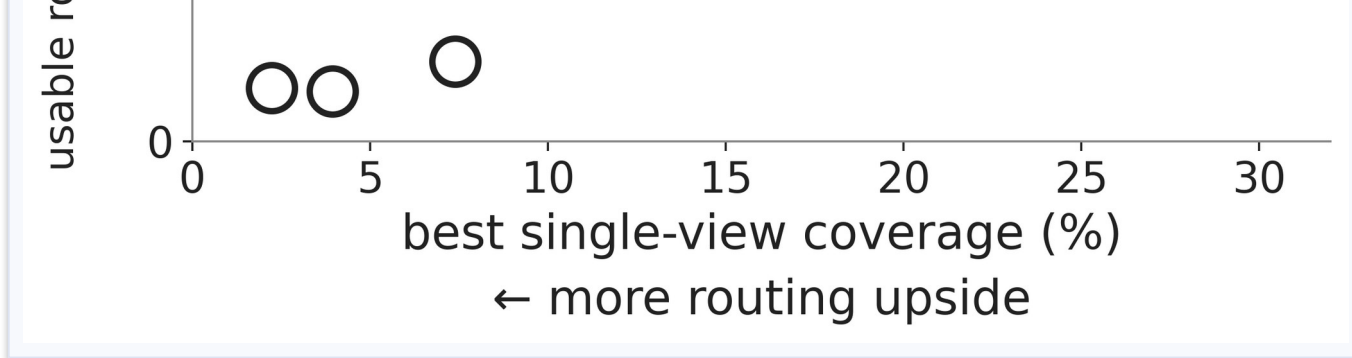
**Learned routers buy success only if
hindsight oracle reaches the win re**

jiaming.wei.25@ucl.ac.uk

Paper, code & full results: github.com/Quarkgluonmixture/Cost-Aware-Routing-for-Web-Usage-Agents



only by spending more; only the win region.



More routing upside, less usable training signal.

[Explore all 8 settings →](#)

Holistic AI × UCL CDI · 16 September 2026

